

Penghua Wang

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SUMMARY

Ph.D. candidate in Data Science and Analytics conducting research on generative AI in healthcare — when LLM and vision-language model outputs are dependable enough for autonomous use in health-related decisions versus requiring human oversight. Lead researcher on an evaluation program for leading commercial and open models, architecting automated LLM-agent pipelines in Python with blinded, human-graded validation and producing decision-ready guidance with pre-specified acceptance criteria and error-rate control; first-author manuscript in preparation. Hands-on generative AI and deep learning development: custom architecture design and training in PyTorch on H100 GPUs, multi-provider LLM API orchestration, and data pipelines for ML on AWS and Azure. First-author publication in Communications Engineering (Nature Portfolio); M.S. in Statistics. Experienced authoring technical reports and presenting AI research to senior leadership, federal agency partners, and cross-functional teams, and collaborating with academic, government, and industry research partners. Available Fall 2026 for a remote internship at 20–40 hours per week.

EDUCATION

Ph.D. Candidate in Data Science and Analytics

University of Oklahoma — Norman, OK — Expected May 2027

Relevant Coursework: Deep Learning, Statistical Machine Learning, Bayesian Statistics, Big Data Analytics, Experimental Design

Master of Science (M.S.) in Statistics

George Washington University — Washington, DC — May 2012

Bachelor of Science (B.S.) in Mathematics

Pittsburg State University — Pittsburg, KS — December 2009

RESEARCH AND PROJECTS

Generative AI Reliability and Safety in Healthcare

Langston University — Oklahoma City, OK

- Lead researcher evaluating leading commercial and open LLMs and vision-language models on real-world health-related tasks, quantifying when AI outputs support autonomous use versus requiring human review — the core question of agentic AI safety and governance
- Architected automated LLM-agent evaluation pipelines in Python that execute judging tasks autonomously, with blinded, human-graded validation confirming automated judgments match human ground truth
- Built the full evaluation stack in Python: API orchestration across LLM providers (including Azure-hosted models), structured-output parsing and validation, automated judging, retry/error handling, and statistical analysis of accuracy and reliability
- Produced decision-ready guidance with pre-specified acceptance criteria and rigorous error-rate control, supporting responsible-AI decisions about automation boundaries in high-stakes health contexts; first-author manuscript in preparation

ML for Healthcare and Biologics Data (Ph.D. Research)

University of Oklahoma / Wheeler Bio — Oklahoma City, OK

- Developed hybrid mechanistic-ML models for multi-step-ahead prediction from longitudinal biological data; benchmarked against random forest, gradient boosting, SVR, and LSTM baselines under rigorous cross-validation
- Achieved ~5.9% mean forecasting error, cutting real assessment timelines from 14–16 days to 9; published as first author in Communications Engineering (Nature Portfolio, 2025)

Deep Learning Architecture Design and Training

University of Oklahoma — Norman, OK

- Designed and trained a custom neural network (PyTorch) for biomedical image interpolation with optical flow and attention mechanisms; trained 20 models on H100 GPUs with mixed precision, outperforming the state-of-the-art baseline

WORK EXPERIENCE

Data Scientist

Langston University — Oklahoma City, OK — March 2021 – Present

- Build data pipelines for ML solutions on large structured and unstructured healthcare and public health datasets: ingestion, cleaning, validation, feature engineering, model development, and deployment
- Communicate AI and analytics findings to senior leadership, federal agency partners, and cross-functional teams; contribute to presentations and technical reports; mentor graduate researchers
- Collaborate with academic research teams, government agencies, and industry partners on ongoing research projects

Visiting Faculty

Florida A&M University — Tallahassee, FL — August 2019 – May 2020

- Taught Business Statistics and Elementary Statistics; mentored undergraduate students

SKILLS

- **Generative AI Research:** LLMs and Vision-Language Models, LLM Evaluation (LLM-as-judge with human-in-the-loop validation), Transformer Architectures, Generative AI Model Development, Prompt Engineering, Retrieval-Augmented Generation (RAG) Concepts, Text Embeddings, Vector Search, Responsible AI, AI Safety Evaluation and AI Governance
- **LLM and Agentic Systems:** Multi-Provider LLM APIs (commercial and open models, Azure-hosted models), LLM-Agent Pipelines, Structured-Output Parsing and Validation, Retry/Error Handling, LangChain (basics), Agentic AI Tools (Claude Code, daily)
- **Machine Learning and Deep Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Custom Architecture Design, Model Development and Validation, NLP, Computer Vision, Time-Series Forecasting, Multi-GPU Training (H100, mixed precision)
- **Data and Cloud:** Python Data Pipelines and ETL for ML, SQL, Large-Scale Structured and Unstructured Healthcare Data, AWS, Azure, Google Cloud Platform, Spark, HPC clusters (Slurm), Docker, Git
- **Statistics:** Hypothesis Testing, Experimental Design, Bayesian Methods, Calibration Analysis, Cross-Validation, Uncertainty Quantification, Error-Rate Control
- **Research and Communication:** Peer-reviewed manuscripts, analytical and technical reports, presentations to senior leadership, academic-industry research collaboration, mentoring graduate researchers

CERTIFICATIONS

SAS Certified Base Programmer — SAS Certified Advanced Programmer — SAS Certified Business Analyst — CFA Level I (passed)

PUBLICATIONS

Wang P., Verma D., Chiu Y., Klier J., Pan C. “Luedeking-Piret regression for multi-step-ahead forecasting and clone selection in monoclonal antibodies biomanufacturing.” Communications Engineering 4, 220 (2025). DOI: <https://doi.org/10.1038/s44172-025-00547-7>